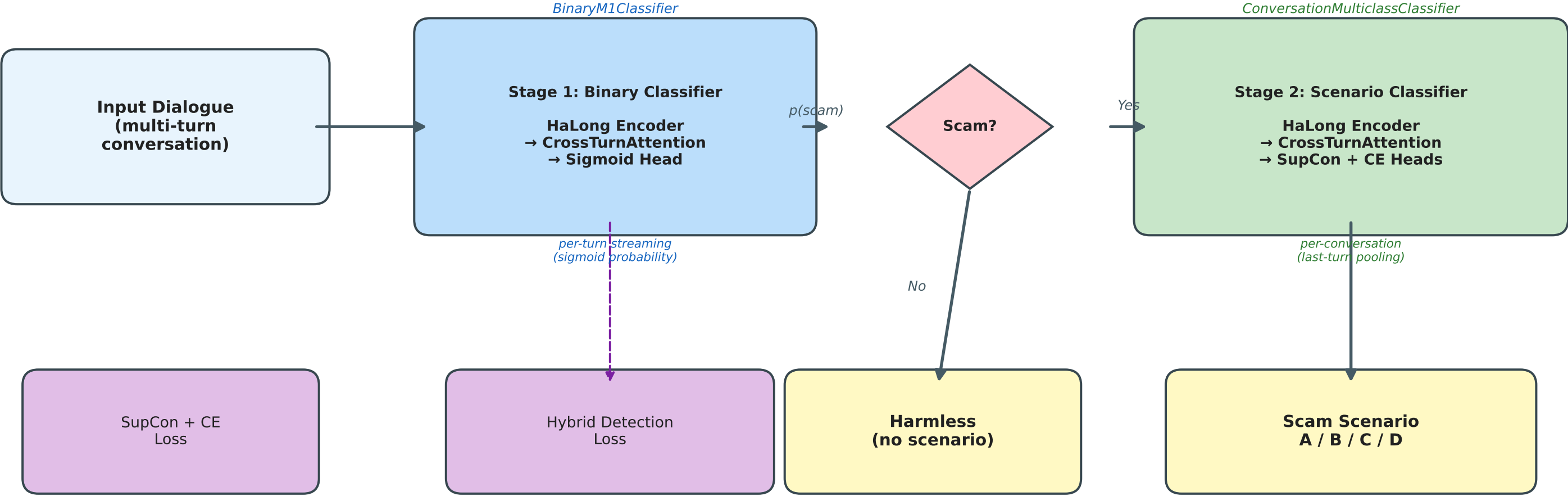


Two-Stage Streaming Scam Detection Pipeline



$$\alpha \cdot SupCon + (1-\alpha) \cdot CE$$

Training: Encoder frozen → full unfreeze at epoch k | AdamW + cosine warmup | gradient checkpointing | truncate augmentation | early stopping